

# Cooking Oils and Combustion Energy

Reading copy of an unpublished working draft by Drish Dedhia. The text and figures below were extracted from the original Word file; this PDF does not reproduce its exact layout. Interpret the findings as provisional.

How does the degree of saturation of fatty acids in different cooking oils (Coconut oil, Sunflower oil, Groundnut oil, Olive oil, and Mustard oil) affect the enthalpy of combustion (measured by a 10 °C temperature rise in water), as determined by calorimetry using a spirit lamp method?

Word count: 2991

## Introduction:

I have been fascinated by cooking shows like the MasterChef franchise and my mother cooking food at home; the different delicacies that she produces from scratch. Although enthralled, I didn't know why one had chosen one type of oil versus another for sautéing or deep-frying, and why foods were cooked faster in some versus others, and even cooked more evenly or flared up at higher temperatures - which hinted at variations in their energy and combustion characteristics. One can just say that each oil tastes differently, but after studying Chemistry in the IB, I learnt to question things at a molecular level. So, using this opportunity in the form of the Chemistry IA, I decided to investigate the enthalpy of combustion of various everyday cooking oils.

## Research Question:

How does the degree of saturation of fatty acids in different cooking oils (Coconut oil, Sunflower oil, Groundnut oil, Olive oil, and Mustard oil) affect the enthalpy of combustion (measured by a 10 °C temperature rise in water), as determined by calorimetry using a spirit lamp method?

## Background:

### Fatty acids:

Fatty acids are energetic molecules in fats, being long hydrocarbon chains terminated with a carboxyl group at one end (as in Figure 1) and are the main fuel in combustion reactions (Smith). Being the simple building blocks of lipids, organic molecules that form the foundation of all life forms, fatty acids are structurally varied. However, it depends on the occurrence and amount of double bonding between carbon atoms in a molecule as this largely affects the energy produced when combustible (Jones).

### Saturation of fatty acids:

The classification of fatty acids is categorized according to the number and position of double bonds along the hydrocarbon chains, which can be either saturated, monounsaturated (MUFA), and polyunsaturated (PUFA) (Meyer, Pan).

**Saturated fatty acids (SFA):** These have only single bonds between carbon atoms, resulting in a straight and rigid structure. This allows them to pack closely together, contributing to higher melting points. Common examples: Lauric acid, Palmitic acid, Stearic acid, etc.

**Monounsaturated fatty acids (MUFA):** These contain one double bond in their carbon chain, introducing a bend that prevents tight packing, leading to lower melting points compared to saturated fats. A prominent example: Oleic

acid.

**Polyunsaturated Fatty Acids (PUFA):** These possess two or more double bonds, resulting in multiple bends in their structure, thus are typically liquid at room temperature and include: Linoleic acid, Alpha-Linolenic acid, etc.

## These acids can be visually represented as in Figure 1:

Figure 1: Representation of fatty acids (Díaz-Puertas et al.)

The degree of saturation in fatty acids affects their physical and chemical properties, including how they interact during combustion.

## Saturation in oils:

Plant oils are mainly triglycerides, which are molecules of glycerol bonded to three fatty acids (Alger). Thus, it can be safely assumed that the saturation level of the fatty acids largely determines the oil's characteristics (Johnson and Baker). which contain 3 long fatty acid chains with a glycerol backbone C<sub>3</sub>H<sub>5</sub> (Wendorrff et al.). This is a generic reaction of triglycerides combusting-



(triglyceride)

Figure 2: Skeletal formula of a triglyceride (B.3 Triglycerides (SL))

R<sub>1</sub>, R<sub>2</sub>, R<sub>3</sub> represent hydrocarbon chain of fatty acids (Wendorrff et al.).

**Highly saturated oils:** Coconut oil and palm oil are saturated with a high amount of saturated fatty acids. These oils have a stable structure, efficiently releasing energy during combustion, thus making them good fuels (Meyer).

**Moderately saturated oils:** MUFA-dominated oils such as olive oil melt at room temperature but can produce much energy when they burn since they contain some saturated elements (Smith).

**Low saturated oil:** PUFA-rich oils are typically liquid. They have lower enthalpy values of combustion than saturated oils. The several double bonds in these oils make them more prone to oxidation, hence less stable, thus they easily combust need produce less energy. (Thompson and Reyes).

Enthalpy of combustion of a substance can be defined as the heat energy released when oxidized completely in the presence of oxygen (BYJU'S). This concept becomes relevant here since cooking oils are composed of triglycerides, which are mainly fatty acids (Wendorrff et al.). The saturated fats burn with more energy because of their stable structure and because there are more carbon-hydrogen bonds available to be oxidized, and this directly correlates with the enthalpy change seen when these oils are burned (Jones). Hence, enthalpy of combustion varies with fatty acid saturation; SFAs release the most energy, then come MUFAs and then PUFAs as double bonds lower the heat released by the substance when it undergoes combustion (Thompson 143).

This study evaluates the effect of fatty acid saturation on enthalpy change by comparing the combustion of coconut, sunflower, olive, mustard and groundnut (peanut) oils. Spirit lamp calorimetry is employed to measure the heat energy transferred to water during combustion. While laboratory constraints prevent direct analysis of fatty acid chain lengths, it is safe to assume the enthalpy value reflects properties of the predominant fatty acid in each oil. By comparing the energy transferred to water as oil burns, it can be determined how the saturation level of fatty acids affects enthalpy and patterns in heat energy between oil types can be investigated (Jones 120).

This research focuses on analysing the effect of saturation on enthalpy change during the combustion of various cooking oils like coconut, sunflower, groundnut, olive, and mustard oil and how does it affect the enthalpy of combustion (measured by a 10 °C temperature rise in water), as determined by calorimetry using a spirit lamp

method.

## Hypothesis:

It is hypothesized that the enthalpy of combustion of cooking oils increases with the degree of saturation in their fatty acid composition, with SFAs showing higher energy release compared to MUFAs or PUFAs.

## Variables:

### Independent Variables:

#### Type of cooking oil:

Coconut oil: Mainly consists of Lauric acid ( $C_{12}H_{24}O_2$ ) - About 45–50% – SFA (NIST Chemistry WebBook; USDA)

Sunflower oil: Primarily composed of Linoleic acid ( $C_{18}H_{34}O_2$ ) - 60–75% - PUFA (NIST Chemistry WebBook; National Library of Medicine)

Groundnut (Peanut) oil: Primarily Oleic acid ( $C_{18}H_{34}O_2$ ) – balanced composition with 40–50% MUFA. (NIST Chemistry WebBook; USDA)

Olive oil: Mainly made of Oleic acid ( $C_{18}H_{34}O_2$ ) - 70–80% - MUFA (NIST Chemistry WebBook; USDA)

Mustard oil: Predominantly Erucic acid ( $C_{22}H_{42}O_2$ ) – MUFA and PUFA (NIST Chemistry WebBook; ScienceDirect)

### Dependent Variables:

Mass of oil required to raise temperature of water by 10°C.

Enthalpy of combustion of different fatty acid compositions.

### Controlled Variables:

| Variable | Rationale for control | Control Method |
|----------|-----------------------|----------------|
|----------|-----------------------|----------------|

|                 |                                                    |                                                  |
|-----------------|----------------------------------------------------|--------------------------------------------------|
| Volume of water | To ensure consistent heat absorption across trials | Use a fixed volume of 100 ml for all experiments |
|-----------------|----------------------------------------------------|--------------------------------------------------|

|                       |                                                                              |                                                       |
|-----------------------|------------------------------------------------------------------------------|-------------------------------------------------------|
| Change in temperature | To measure consistent amount of heating for all oils within a suitable range | Fix the temperature change to be 10 °C for each trial |
|-----------------------|------------------------------------------------------------------------------|-------------------------------------------------------|

|                          |                                                          |                                                                |
|--------------------------|----------------------------------------------------------|----------------------------------------------------------------|
| Environmental conditions | To minimize external influences on combustion efficiency | Conduct all experiments in a controlled laboratory environment |
|--------------------------|----------------------------------------------------------|----------------------------------------------------------------|

|                     |                                                              |                                                               |
|---------------------|--------------------------------------------------------------|---------------------------------------------------------------|
| Type of spirit lamp | To ensure consistent fuel source and burning characteristics | Use the same spirit with repeated washing lamp for all trials |
|---------------------|--------------------------------------------------------------|---------------------------------------------------------------|

Table 1: Controlled variables

## Apparatus:

Spirit lamp x 1

Cotton - 50 g

100 ml Copper calorimeter x 1

Retort stand x 1

Retort stand clamp x 1

Wind shield x 1

Thermometer ( $\pm 0.1^{\circ}\text{C}$ ) x 1

Measuring cylinder ( $100 \pm 1$  ml) x 1

Mass balance ( $\pm 0.01$  g) x 1

Lighter x 1

Safety goggles and gloves

## Methodology:

### Experimental design and rationale:

This investigation used the conventional method of spirit lamp calorimetry but with oils in the spirit lamp instead of alcohols, to obtain combustion enthalpy. This was selected because since it is simple, reproducible, applicable in a school laboratory and appropriate for quantifying enthalpy change for comparing. Nonetheless problems arose with oils in its viscosity and density, which restricted capillary action, disabling the oil to flow through the wick to the flame.

To overcome this, standard rope wicks were replaced with cotton wicks which were handmade to fit the spirit lamp. Cotton was used because it has low density, burns fast, and supports capillary action through which oils could easily pass to reach the flame efficiently and maintain constant burning rate. All the wicks were sized equally for consistency in trials.

During the pilot readings, it was observed that the wick extinguished quickly due to the custom cotton wick burning fast, preventing the water from having a large increase in temperature. To ensure consistent and comparable data for all trials, the temperature change ( $\Delta T$ ) was standardized to a small value of  $10^{\circ}\text{C}$  per trial.

This modified technique allowed oils to burn smoothly and ensured data was appropriately collected. Mass changes for all five attempts in each of the five oils were used for calculating enthalpy of combustion of each oil, hence enabling comparison.

### Experimental set-up:

Take some cotton and roll it up to form a wick.

Replace the original wick of the spirit lamp with this cotton wick.

Fill a dry, empty spirit lamp with sunflower oil.

Dip the wick in oil ensuring it is completely soaked.

Insert it through the lamp's cap and securely fasten the cap back.

Suspend the copper calorimeter vessel above the spirit lamp using a retort stand clamp and the retort stand as shown in Figure 3.

Attach a thermometer vertically above the vessel on a retort stand with another clamp such that its bulb is inside the vessel as shown in Figure 3.

Place the wind shield around the set-up as shown in Figure 4.

Figure 3: Set-up

Figure 4: Wind shield placement

## Experimental procedure:

Measure and record the mass of the spirit lamp with its cap using the mass balance as shown in Figure 4 below.

Measure 100 ml of water using a measuring cylinder and pour it into the vessel.

Record the initial temperature of water.

Light the wick using a lighter.

Wait until the desired temperature change of +10°C is reached.

Extinguish the flame, measure the final temperature.

Measure and record the final mass of the spirit lamp with its cap.

Repeat steps 1 through 7 for 5 trials each of all 5 oils.

Figure 5: Measuring mass of spirit lamp

Figure 6: Top view of experiment

Figure 7: Side view of experiment

## Safety Precautions:

Safety Precaution | Rationale | Implementation

Wear lab coat, safety goggles, and heat-resistant gloves | Protects skin and eyes from hot oil, flame, and potential splashes. | Ensure all safety gear is worn properly before the experiment is conducted.

Work in a well-ventilated area | Burning oils can produce fumes that may leave a lingering smell in the room and could also irritate the lungs | Conduct the experiment in a properly ventilated chemistry lab with the exhaust fans switched on.

| Skin may react to specific oils. | Ensure to use oils to which you aren't allergic, wash hands thoroughly after handling each oil and wear gloves to be safe.

Stable surface and secure apparatus | Ensures that equipment remains steady, reducing risk of spills or accidental knocks. | Use clamps and stands to secure equipment, and check stability before lighting the flame.

Avoid excessive oil in the spirit lamp | Reduces the chance of large flames and uncontrolled burning and decrease wastage of oils. | Add oil in small quantities, just enough for 5 trials.

Fire hazard | Unattended flame may lead to a fire. | Attend to the flame at all times.

Table 2: Safety precautions

## Results:

## Raw Data:

Raw Data Table | Raw Data Table | Raw Data Table | Raw Data Table | Raw Data Table | Raw Data Table

Raw Data Table | Raw Data Table | Raw Data Table | Raw Data Table | Raw Data Table | Raw Data Table |

Oil Type; (Formula); (Molar mass) (IB Chemistry Data Booklet) | Trial no. |  $M_i$ ; (Initial mass) (g) |  $M_f$ ; (Final mass) (g) |  $T_i$  (Initial temperature); ( $^{\circ}\text{C}$ ) |  $T_f$  (Final temperature); ( $^{\circ}\text{C}$ ) |

|                                                                                    |   |        |        |    |    |
|------------------------------------------------------------------------------------|---|--------|--------|----|----|
| COCONUT; ( $\text{C}_{33}\text{H}_{56}\text{O}_2$ ); ; Molar mass- 200.32 g/mol    | 1 | 98.88  | 98.66  | 30 | 40 |
| COCONUT; ( $\text{C}_{33}\text{H}_{56}\text{O}_2$ ); ; Molar mass- 200.32 g/mol    | 2 | 99.02  | 98.78  | 29 | 39 |
| COCONUT; ( $\text{C}_{33}\text{H}_{56}\text{O}_2$ ); ; Molar mass- 200.32 g/mol    | 3 | 99.54  | 99.35  | 29 | 39 |
| COCONUT; ( $\text{C}_{33}\text{H}_{56}\text{O}_2$ ); ; Molar mass- 200.32 g/mol    | 4 | 100.36 | 100.13 | 29 | 39 |
| COCONUT; ( $\text{C}_{33}\text{H}_{56}\text{O}_2$ ); ; Molar mass- 200.32 g/mol    | 5 | 100.60 | 100.35 | 30 | 40 |
| SUNFLOWER; ( $\text{C}_{33}\text{H}_{56}\text{O}_2$ ); ; Molar mass-; 280.45 g/mol | 1 | 104.85 | 104.48 | 29 | 39 |
| SUNFLOWER; ( $\text{C}_{33}\text{H}_{56}\text{O}_2$ ); ; Molar mass-; 280.45 g/mol | 2 | 105.23 | 104.92 | 29 | 39 |
| SUNFLOWER; ( $\text{C}_{33}\text{H}_{56}\text{O}_2$ ); ; Molar mass-; 280.45 g/mol | 3 | 105.47 | 105.19 | 29 | 39 |
| SUNFLOWER; ( $\text{C}_{33}\text{H}_{56}\text{O}_2$ ); ; Molar mass-; 280.45 g/mol | 4 | 105.52 | 105.22 | 29 | 39 |
| SUNFLOWER; ( $\text{C}_{33}\text{H}_{56}\text{O}_2$ ); ; Molar mass-; 280.45 g/mol | 5 | 106.09 | 105.77 | 29 | 39 |
| GROUNDNUT; ( $\text{C}_{33}\text{H}_{56}\text{O}_2$ ); ; Molar mass-; 282.47 g/mol | 1 | 110.13 | 109.87 | 26 | 36 |
| GROUNDNUT; ( $\text{C}_{33}\text{H}_{56}\text{O}_2$ ); ; Molar mass-; 282.47 g/mol | 2 | 110.16 | 109.84 | 30 | 40 |
| GROUNDNUT; ( $\text{C}_{33}\text{H}_{56}\text{O}_2$ ); ; Molar mass-; 282.47 g/mol | 3 | 110.21 | 109.79 | 30 | 40 |
| GROUNDNUT; ( $\text{C}_{33}\text{H}_{56}\text{O}_2$ ); ; Molar mass-; 282.47 g/mol | 4 | 109.93 | 109.54 | 30 | 40 |
| GROUNDNUT; ( $\text{C}_{33}\text{H}_{56}\text{O}_2$ ); ; Molar mass-; 282.47 g/mol | 5 | 110.44 | 110.11 | 29 | 39 |
| OLIVE; ( $\text{C}_{33}\text{H}_{56}\text{O}_2$ ); ; Molar mass - 282.47 g/mol     | 1 | 89.19  | 88.82  | 28 | 38 |
| OLIVE; ( $\text{C}_{33}\text{H}_{56}\text{O}_2$ ); ; Molar mass - 282.47 g/mol     | 2 | 89.11  | 88.71  | 28 | 38 |
| OLIVE; ( $\text{C}_{33}\text{H}_{56}\text{O}_2$ ); ; Molar mass - 282.47 g/mol     | 3 | 89.32  | 89.03  | 28 | 38 |
| OLIVE; ( $\text{C}_{33}\text{H}_{56}\text{O}_2$ ); ; Molar mass - 282.47 g/mol     | 4 | 89.29  | 88.91  | 27 | 37 |
| OLIVE; ( $\text{C}_{33}\text{H}_{56}\text{O}_2$ ); ; Molar mass - 282.47 g/mol     | 5 | 89.49  | 89.10  | 27 | 37 |
| MUSTARD; ( $\text{C}_{33}\text{H}_{56}\text{O}_2$ ); ; Molar mass –; 338.58 g/mol  | 1 | 122.75 | 122.38 | 29 | 39 |
| MUSTARD; ( $\text{C}_{33}\text{H}_{56}\text{O}_2$ ); ; Molar mass –; 338.58 g/mol  | 2 | 122.77 | 122.37 | 29 | 39 |
| MUSTARD; ( $\text{C}_{33}\text{H}_{56}\text{O}_2$ ); ; Molar mass –; 338.58 g/mol  | 3 | 122.88 | 122.45 | 29 | 39 |
| MUSTARD; ( $\text{C}_{33}\text{H}_{56}\text{O}_2$ ); ; Molar mass –; 338.58 g/mol  | 4 | 122.95 | 122.53 | 28 | 38 |
| MUSTARD; ( $\text{C}_{33}\text{H}_{56}\text{O}_2$ ); ; Molar mass –; 338.58 g/mol  | 5 | 123.21 | 122.76 | 29 | 39 |

Table 3: Raw Data table

## Sample calculation:

**Example taken here is the last trial of Mustard oil given in the table:**

$\Delta T$  (Change in temperature):  $T_f - T_i = 39 - 29 = 10\text{ }^{\circ}\text{C}$

(Mass of oil combusted = change in mass):  $M_i - M_f$  (as mass decreases and mass can't be negative) =  $123.21 - 122.76 = 0.45\text{ g}$

(Heat energy transferred to water), where m is the mass of water, c is the specific heat capacity of water and  $\Delta T$  is the change in temperature of water

$$= 100\text{g} \times 4.18 \text{ J g}^{-1} \times 10 = 4180 \text{ J}$$

(Note: negative sign in Q is ignored since it just shows that heat is released in the reaction.) (100 g of water is used as 100 ml of water was heated and density of water was assumed to be  $1 \text{ g cm}^{-3} = 1 \text{ g ml}^{-1}$ )

$$\text{Moles of oil used (n)} = 0.001329 \text{ mol}$$

$$\text{Enthalpy of combustion} = 4180 / (0.001329) = 3145032.00 \text{ J mol}^{-1}$$

$$\text{Enthalpy in KJ} = 3145.03 \text{ KJ mol}^{-1}$$

$$\text{Average enthalpy} = 3433.84 \text{ KJ mol}^{-1}$$

## Processed Data:

Oil Type | Change in mass (g) | Molar mass of oil - g/mol (Mr) | Moles of oil - mol (m/Mr) | Enthalpy - delta H (Q/n) (J/mol) | Enthalpy in KJ/mol | Uncertainty ( $\pm$ ) (KJ mol<sup>-1</sup>) | Average value of enthalpy of combustion KJ/mol | Uncertainty (KJ mol<sup>-1</sup>)

|           |      |        |          |            |         |            |         |                  |
|-----------|------|--------|----------|------------|---------|------------|---------|------------------|
| COCONUT   | 0.22 | 200.32 | 0.001098 | 3806080.00 | 3806.08 | 0.00009091 | 3865.99 | $\pm 0.00046170$ |
| COCONUT   | 0.24 | 200.32 | 0.001198 | 3488906.67 | 3488.91 | 0.00008333 | 3865.99 | $\pm 0.00046170$ |
| COCONUT   | 0.19 | 200.32 | 0.000948 | 4407040.00 | 4407.04 | 0.00010526 | 3865.99 | $\pm 0.00046170$ |
| COCONUT   | 0.23 | 200.32 | 0.001148 | 3640598.26 | 3640.60 | 0.00008696 | 3865.99 | $\pm 0.00046170$ |
| COCONUT   | 0.21 | 200.32 | 0.001048 | 3987321.90 | 3987.32 | 0.00009524 | 3865.99 | $\pm 0.00046170$ |
| SUNFLOWER | 0.37 | 280.45 | 0.001319 | 3168327.03 | 3168.33 | 0.00005405 | 3741.52 | $\pm 0.00031916$ |
| SUNFLOWER | 0.31 | 280.45 | 0.001105 | 3781551.61 | 3781.55 | 0.00006452 | 3741.52 | $\pm 0.00031916$ |
| SUNFLOWER | 0.28 | 280.45 | 0.000998 | 4186717.86 | 4186.72 | 0.00007143 | 3741.52 | $\pm 0.00031916$ |
| SUNFLOWER | 0.30 | 280.45 | 0.001070 | 3907603.33 | 3907.60 | 0.00006667 | 3741.52 | $\pm 0.00031916$ |
| SUNFLOWER | 0.32 | 280.45 | 0.001141 | 3663378.12 | 3663.38 | 0.00006272 | 3741.52 | $\pm 0.00031916$ |
| GROUNDNUT | 0.26 | 282.47 | 0.000920 | 4541248.46 | 4541.25 | 0.00007692 | 3529.54 | $\pm 0.00029893$ |
| GROUNDNUT | 0.32 | 282.47 | 0.001133 | 3689764.38 | 3689.76 | 0.00006250 | 3529.54 | $\pm 0.00029893$ |
| GROUNDNUT | 0.42 | 282.47 | 0.001487 | 2811249.05 | 2811.25 | 0.00004762 | 3529.54 | $\pm 0.00029893$ |
| GROUNDNUT | 0.39 | 282.47 | 0.001381 | 3027498.97 | 3027.50 | 0.00005128 | 3529.54 | $\pm 0.00029893$ |
| GROUNDNUT | 0.33 | 282.47 | 0.001168 | 3577953.33 | 3577.95 | 0.00006061 | 3529.54 | $\pm 0.00029893$ |
| OLIVE     | 0.37 | 282.47 | 0.001310 | 3191147.57 | 3191.15 | 0.00005405 | 3269.82 | $\pm 0.00027693$ |
| OLIVE     | 0.40 | 282.47 | 0.001416 | 2951811.50 | 2951.81 | 0.00005000 | 3269.82 | $\pm 0.00027693$ |
| OLIVE     | 0.29 | 282.47 | 0.001027 | 4071464.14 | 4071.46 | 0.00006897 | 3269.82 | $\pm 0.00027693$ |
| OLIVE     | 0.38 | 282.47 | 0.001345 | 3107170.00 | 3107.17 | 0.00005263 | 3269.82 | $\pm 0.00027693$ |
| OLIVE     | 0.39 | 282.47 | 0.001381 | 3027498.97 | 3027.50 | 0.00005128 | 3269.82 | $\pm 0.00027693$ |
| MUSTARD   | 0.37 | 338.58 | 0.001093 | 3825038.92 | 3825.04 | 0.00005405 | 3433.84 | $\pm 0.00024263$ |
| MUSTARD   | 0.40 | 338.58 | 0.001181 | 3538161.00 | 3538.16 | 0.00005000 | 3433.84 | $\pm 0.00024263$ |
| MUSTARD   | 0.43 | 338.58 | 0.001270 | 3291312.56 | 3291.31 | 0.00004651 | 3433.84 | $\pm 0.00024263$ |
| MUSTARD   | 0.42 | 338.58 | 0.001240 | 3369677.14 | 3369.68 | 0.00004762 | 3433.84 | $\pm 0.00024263$ |
| MUSTARD   | 0.45 | 338.58 | 0.001329 | 3145032.00 | 3145.03 | 0.00004444 | 3433.84 | $\pm 0.00024263$ |

Table 4: Processed data table

Energy change =  $Q = 4180 \text{ J}$

Uncertainty for Average value of combustion =

(Since literature values from the IB Chemistry Data booklet and other cited sources have been used, uncertainty is ignored.)

Graph 1: Representation of experimental data

(Error bars are included on the graph but are barely visible due to their small size)

## Analysis:

As Graph 1 rightly summarizes, the highest enthalpy of combustion calculated was of coconut oil as hypothesised since it has high SFA content. Then it is followed by sunflower oil which contains predominantly PUFAs, then by groundnut oil with a balanced combination of saturated and unsaturated fatty acids, then by mustard oil with both unsaturated fatty acid content being greater, and finally olive oil, with mainly MUFA content.

To check results' reliability, a weighted saturation index was used, that measures the total sum of fatty acid saturation in cooking oils. This index takes into consideration the relative proportion of saturated (SFA), monounsaturated (MUFA), and polyunsaturated fatty acids (PUFA). SFAs are weighted the highest and are  $\times 3$  due to complete saturation, MUFAs receive an intermediate weighting at  $\times 2$  due to partial unsaturation. PUFAs are, of course, weighted low at  $\times 1$  due to multiple unsaturated bonds in the fatty acid chain. This weighting system provides numerical representation of each oil's fatty acid composition, whereby a more precise correlation can be made with measured enthalpies of combustion. The index values themselves are positive numbers and therefore directly reflect the general saturation character of each oil as having higher values corresponding to greater degrees of saturation.

| Oil | SFA (%) | MUFA (%) | PUFA (%) |
|-----|---------|----------|----------|
|-----|---------|----------|----------|

|         |    |   |   |
|---------|----|---|---|
| Coconut | 91 | 8 | 1 |
|---------|----|---|---|

|           |    |    |    |
|-----------|----|----|----|
| Sunflower | 11 | 23 | 66 |
|-----------|----|----|----|

|           |    |    |    |
|-----------|----|----|----|
| Groundnut | 18 | 53 | 29 |
|-----------|----|----|----|

|       |    |    |    |
|-------|----|----|----|
| Olive | 14 | 76 | 10 |
|-------|----|----|----|

|         |   |    |    |
|---------|---|----|----|
| Mustard | 7 | 72 | 21 |
|---------|---|----|----|

Table 5: SFA, MUFA and PUFA content percentage (Sharma and Kanwaria)

## Formula for saturation index:

| Oil | Saturation Index | Average value of enthalpy of combustion KJ/mol |
|-----|------------------|------------------------------------------------|
|-----|------------------|------------------------------------------------|

|         |     |         |
|---------|-----|---------|
| Coconut | 2.9 | 3865.99 |
|---------|-----|---------|

|           |      |         |
|-----------|------|---------|
| Sunflower | 1.45 | 3741.52 |
|-----------|------|---------|

|           |      |         |
|-----------|------|---------|
| Groundnut | 1.89 | 3529.54 |
|-----------|------|---------|

|       |      |         |
|-------|------|---------|
| Olive | 2.04 | 3269.82 |
|-------|------|---------|

|         |      |         |
|---------|------|---------|
| Mustard | 1.86 | 3433.84 |
|---------|------|---------|

Table 6: Saturation index

Saturation index values usually fall within the range of 1.0 to 3.0, where values closer to 3.0 indicate high saturation, about 2.0 imply partial saturation (balanced mixture or mostly MUFA) and values closer to 1.0 indicate low saturation and high levels of PUFA.

Certain observations that can be identified through the data are as follows:

The highest saturation index is found to be of coconut oil, at 2.90, corresponding with the highest enthalpy of combustion, at 3865.99 kJ/mol. This indicates that an increased saturation may have positive relationship with higher energy content.

This, however, is not linearly related, as demonstrated by Sunflower oil that, despite having a saturation index of 1.45, shows the second-highest enthalpy value at 3741.52 kJ/mol.

The trend generally aligns with the findings from the experiment, however the outlier values, indicate that the relationship between fatty acid composition and enthalpy of combustion must be complex and depend on yet other factors except saturation of fatty acids in triglycerides that are not captured in this analysis. This can be a good extension of research for under this topic.

## Errors and Uncertainties:

### Error:

The density of oil and capillary action in the wick determine the flow of oil through the wick, which causes fluctuations in the burn rate and flame size, directly affecting the enthalpy of combustion measurements.

In recording the initial and final mass of oil, and in the measured temperature change of water, small differences can be enough to affect how accurately enthalpy measurements per gram or mole would become - especially when scaled up.

Variability in flame height, often due to impurities in mixed composition oils like groundnut oil, leads to flickering or irregular flame height and therefore a change in the heat produced and the resultant enthalpy values.

There may be heat loss when some of the thermal energy escapes to the surroundings rather than being completely transferred to the water, especially for higher temperature flames, this can be affected further by differences in calorimeter insulation and environmental conditions.

Parallax errors in the reading of the thermometer may cause slight inaccuracies in the recorded temperature values.

### Therefore, final results:

Oil | Average value of enthalpy of combustion (KJ mol<sup>-1</sup>) | Absolute uncertainty of average enthalpy of combustion (±) (KJ mol<sup>-1</sup>)  
| Percentage uncertainty; (%)

Coconut | 3865.99 | 0.0004617 | 0.00001194

Sunflower | 3741.52 | 0.00031917 | 0.00000853

Groundnut | 3529.54 | 0.00029893 | 0.00000847

Olive | 3269.82 | 0.00027693 | 0.00000847

Mustard | 3433.84 | 0.00024263 | 0.00000707

Table 7: Results with uncertainty

## Evaluation:

### Limitations and improvements

The assumption that fatty acid saturation primarily influences the enthalpy of combustion is overly broad and overlooks the complex composition of oils and other components beyond fatty acids.

Commercial oil impurities affecting enthalpy: Cooking oils are not lab-grade and hence may be carrying impurities, thereby affecting combustion energy. Lab-grade oils will yield a better accuracy (Smith and Jones).

Heat loss to surroundings despite insulation: Some heat inevitably escaped during combustion. Using a bomb calorimeter would better contain heat.

Inconsistent wick absorption rates: Custom wicks may have minor differences in size and shape, leading to inconsistency in capillary action. Other better conducting wicks can be used.

Parallax error in manual temperature readings: Human error impacted temperature accuracy. Digital thermometers would yield consistent readings.

Assumed constant specific heat capacity of water: Minor temperature changes in water's specific heat capacity were overlooked. Adjusting for this would improve precision.

## Extensions for Further Research:

Testing in oxygen-enriched environments would provide peak enthalpy values for biofuel potential (Stevens et al.).

A comparative biofuel study with synthetics would clearly outline the benefits of sustainability by comparing vegetable oils to synthetic biofuels.

Fatty acid composition blending analysis: Testing of blended oils could optimize biofuel formulations by type of fatty acid.

Thermogravimetric analysis: This would give decomposition data and thus a better insight into how well each oil burns (Smith and Jones).

## Conclusion:

The results of this study validate the hypothesis suggesting that the combustion enthalpy of cooking oils increases with the degree of saturation in their fatty acid composition. The enthalpy of combustion for each of the type of oil namely coconut, sunflower, groundnut, olive and mustard present a clear descending order in values consistent with the respective levels of saturation of the oils. Oils high in saturated fatty acids (SFAs), namely, coconut oil had the highest enthalpy of combustion  $3865.99 \text{ KJ mol}^{-1}$  while others high in unsaturation like olive oil had very low values  $3269.82 \text{ KJ mol}^{-1}$ . This implies that the energy that comes out from these oils is strongly associated with inter-molecular structure, thus this observed trend is consistent with the theoretical expectations. The alignment of experimental value trend with the trend in saturation indexes along with the small amount of uncertainty, both absolute and percentage, adds to the reliability of the experiment, however there are limitations to the methodology as mentioned above. Hence it can be safely concluded that higher degree of saturation of fatty acids, as in triglycerides in cooking oils, leads to higher enthalpy of combustion.

### Works Cited

Alger, Angela Harter. "6.1 Triglycerides and Fatty Acids." Calstate.edu, Pressbooks, 2022, [pressbooks.calstate.edu/nutritionandfitness/chapter/6-1-triglycerides-and-fatty-acids/](https://pressbooks.calstate.edu/nutritionandfitness/chapter/6-1-triglycerides-and-fatty-acids/). Accessed 14 Jan. 2025.

"An Intricate Review on Nutritional and Analytical Profiling of Coconut, Flaxseed, Olive, and Sunflower Oil Blends." MDPI, 2024.

B.3 Triglycerides (SL). "B.3 Triglycerides (SL)." YouTube, 9 June 2015, [www.youtube.com/watch?v=KtPbwXolNag&t=53s](https://www.youtube.com/watch?v=KtPbwXolNag&t=53s). Accessed 24 Dec. 2024.

BYJU'S. "Define Enthalpy of Combustion-." Byjus.com, BYJU'S, 4 July 2022, [byjus.com/question-answer/define-enthalpy-of-combustion/](https://byjus.com/question-answer/define-enthalpy-of-combustion/). Accessed 24 Dec. 2024.

C, Zambiazzi R., et al. "Fatty Acid Composition of Vegetable Oils and Fats." B. CEPPA Curitiba, vol. 25, 2007, pp. 111–20.

Chemistry WebBook. National Institute of Standards and Technology (NIST), 2023.

Coconut Oil. National Center for Biotechnology Information (NCBI), 2024.

Díaz-Puertas, Rocío, et al. "Fish Skin Mucus Extracts: An Underexplored Source of Antimicrobial Agents." *Marine Drugs*, vol. 21, June 2023, p. 350, <https://doi.org/10.3390/md21060350>.

Erucic Acid. NIST Chemistry WebBook, 2024.

G, Gillingham L. "Dietary Monounsaturated Fatty Acids Are Protective against Metabolic Syndrome and Cardiovascular Disease Risk Factors." National Center for Biotechnology Information, 2024.

Johnson, Emily, and Patrick Baker. *Chemistry of Fats and Oils*. Oxford UP, 2018.

Jones, Sarah. *Calorimetry Fundamentals in Chemistry*. Springer, 2019.

Kostik, V., et al. "Fatty Acid Composition of Edible Oils and Fats." *Journal of Hygienic Engineering and Design*, vol. 4, 2013, pp. 112–16.

Lauric Acid. NIST Chemistry WebBook, 2024.

Linoleic Acid. NIST Chemistry WebBook, 2024.

Meyer, Lucas. "Fatty Acid Chemistry and Energy." *Journal of Bioenergetics*, vol. 45, 2, 2020, pp. 57–65.

Mustard Oil Composition. ScienceDirect, 2024.

Oleic Acid. NIST Chemistry WebBook, 2024.

Olive Oil. USDA FoodData Central, 2024.

Pan, Sourav. "Lipids - Definition, Structure, Properties, Types, Functions, Examples - Biology Notes Online." *Biology Notes Online*, 31 Mar. 2024, [biologynotesonline.com/lipids/](https://biologynotesonline.com/lipids/). Accessed 30 Nov. 2024.

Peanut Oil Composition. USDA FoodData Central, 2024.

Sharma, Rashmi, and Shekhar Kanwaria. "Comparative Chemistry of Mustard, Groundnut, Sesame, Soyabean and Sunflower Oils and Impact on Human Health | Biores Scientia." *Biores Scientia*, 2023, [bioresscientia.com/article/comparative-chemistry-of-mustard-groundnut-sesame-soyabean-and-sunflower-oils-and-impact-on-human-health](https://bioresscientia.com/article/comparative-chemistry-of-mustard-groundnut-sesame-soyabean-and-sunflower-oils-and-impact-on-human-health). Accessed 23 Nov. 2024.

Smith, Alicia. *Organic Chemistry of Lipids*. HarperCollins, 2017.

Smith, John, and Emily Jones. "Analyzing Impurities in Commercial Oils." *Journal of Organic Chemistry*, 2023.

Stevens, Karen. "Heat Loss and Flame Stability in Vegetable Oil Combustion." *International Journal of Chemical Sciences*, vol. 12, 2, 2023.

Sunflower Oil. National Library of Medicine, 2024.

Thompson, Gregory, and Lucia Reyes. "Polyunsaturated Fats in Combustion." *Fuel Science Journal*, vol. 52, 1, 2021, pp. 139–50.

Wendorrff, Tim, et al. "3.5.1: Foods- Burning or Metabolizing Fats and Sugars." Chemistry LibreTexts, 13 May 2016, [chem.libretexts.org/Bookshelves/General\\_Chemistry/ChemPRIME\\_%28Moore\\_et\\_al.%29/03%3A\\_Using\\_Chemical\\_Equations\\_in\\_Calculations/3.05%3A\\_Analysis\\_of\\_Compounds/3.5.01%3A\\_Foods-\\_Burning\\_or\\_Metabolizing\\_Fats\\_and\\_Sugars?utm\\_source=chatgpt.com](https://chem.libretexts.org/Bookshelves/General_Chemistry/ChemPRIME_%28Moore_et_al.%29/03%3A_Using_Chemical_Equations_in_Calculations/3.05%3A_Analysis_of_Compounds/3.5.01%3A_Foods-_Burning_or_Metabolizing_Fats_and_Sugars?utm_source=chatgpt.com). Accessed 24 Dec. 2024.

## Figures from the working draft

image1.png

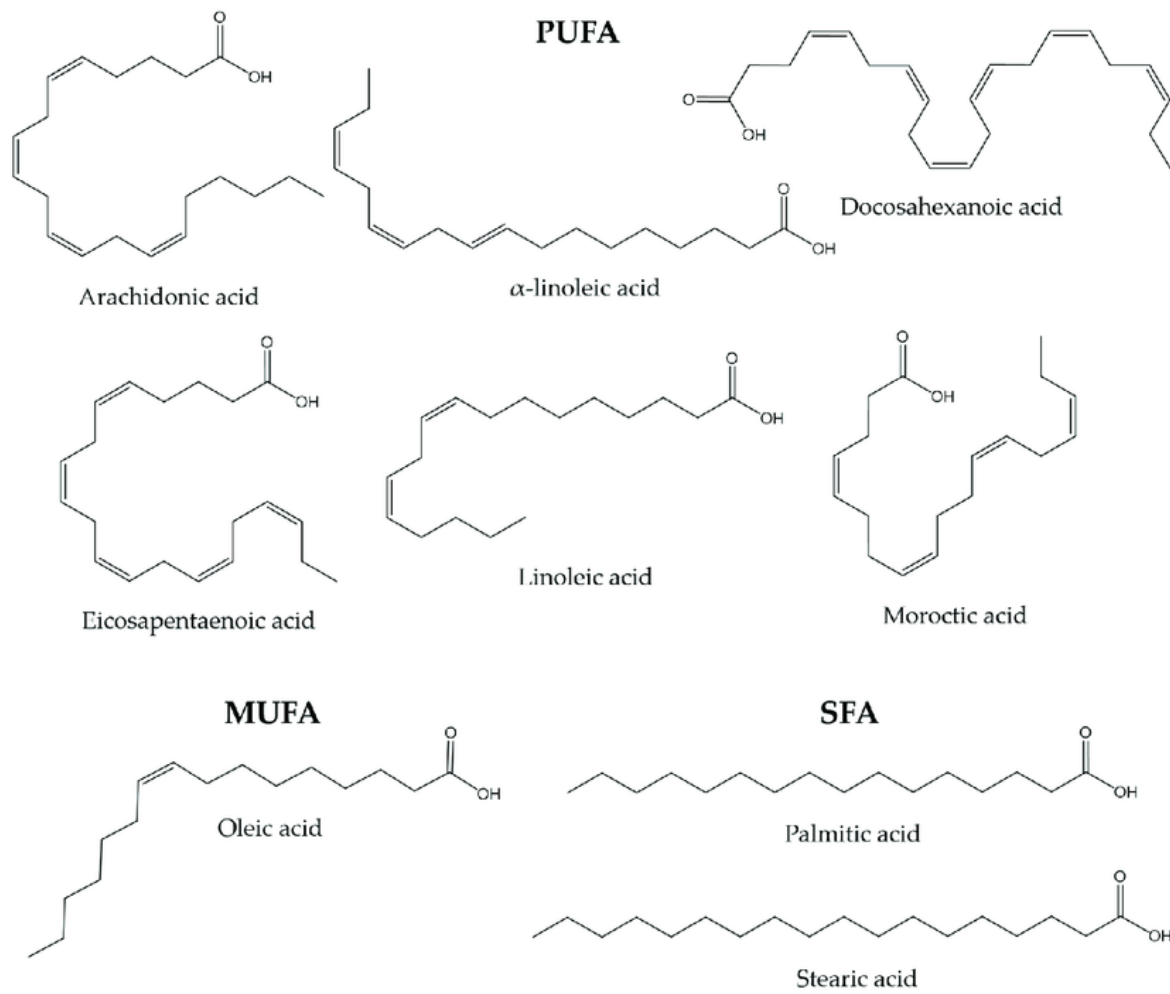


image2.png

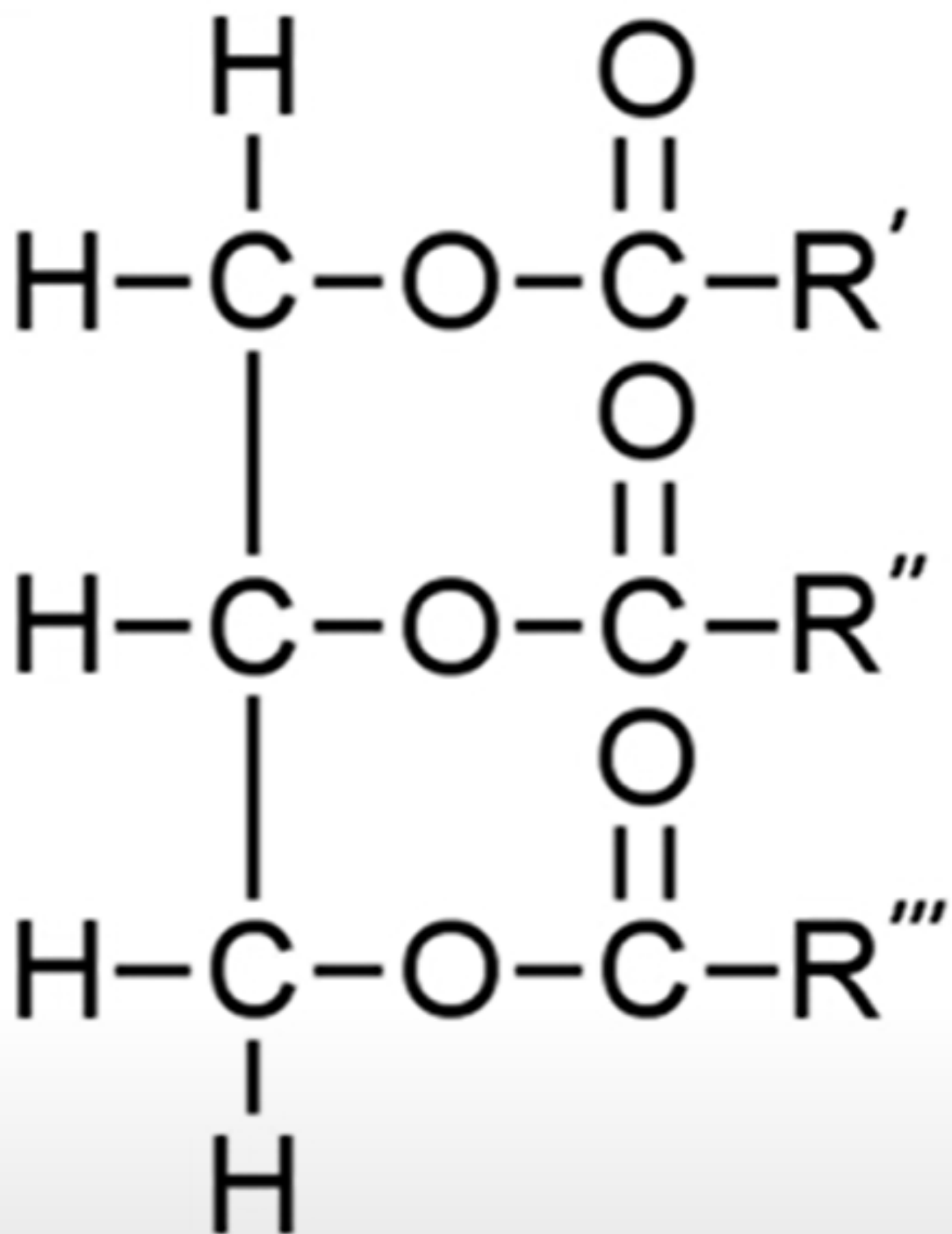


image3.png

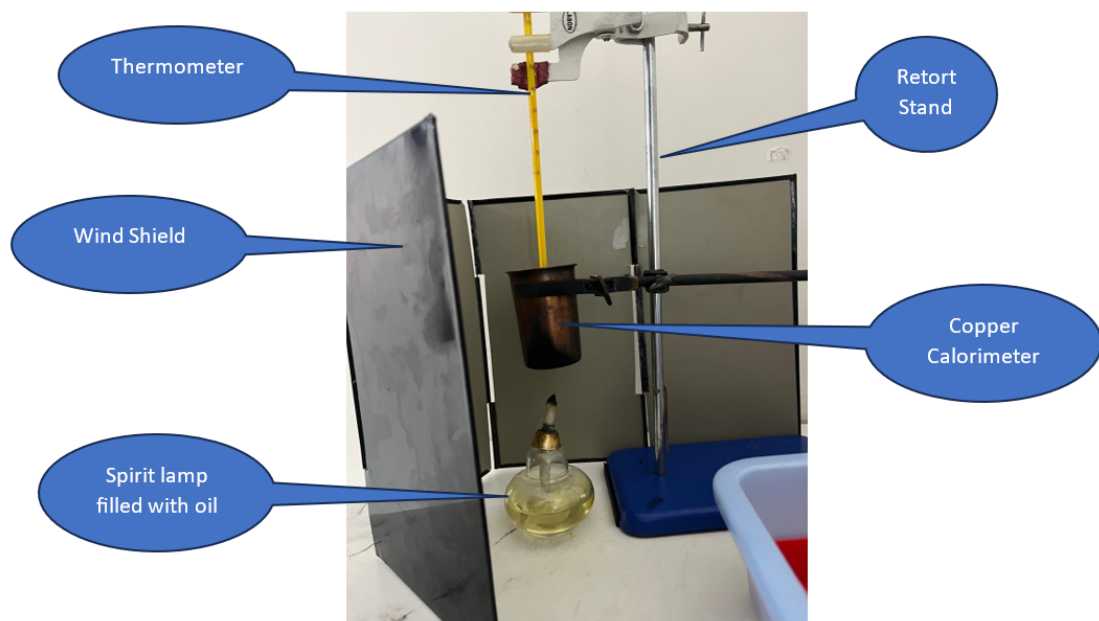


image4.jpeg



image5.jpeg



image6.jpeg

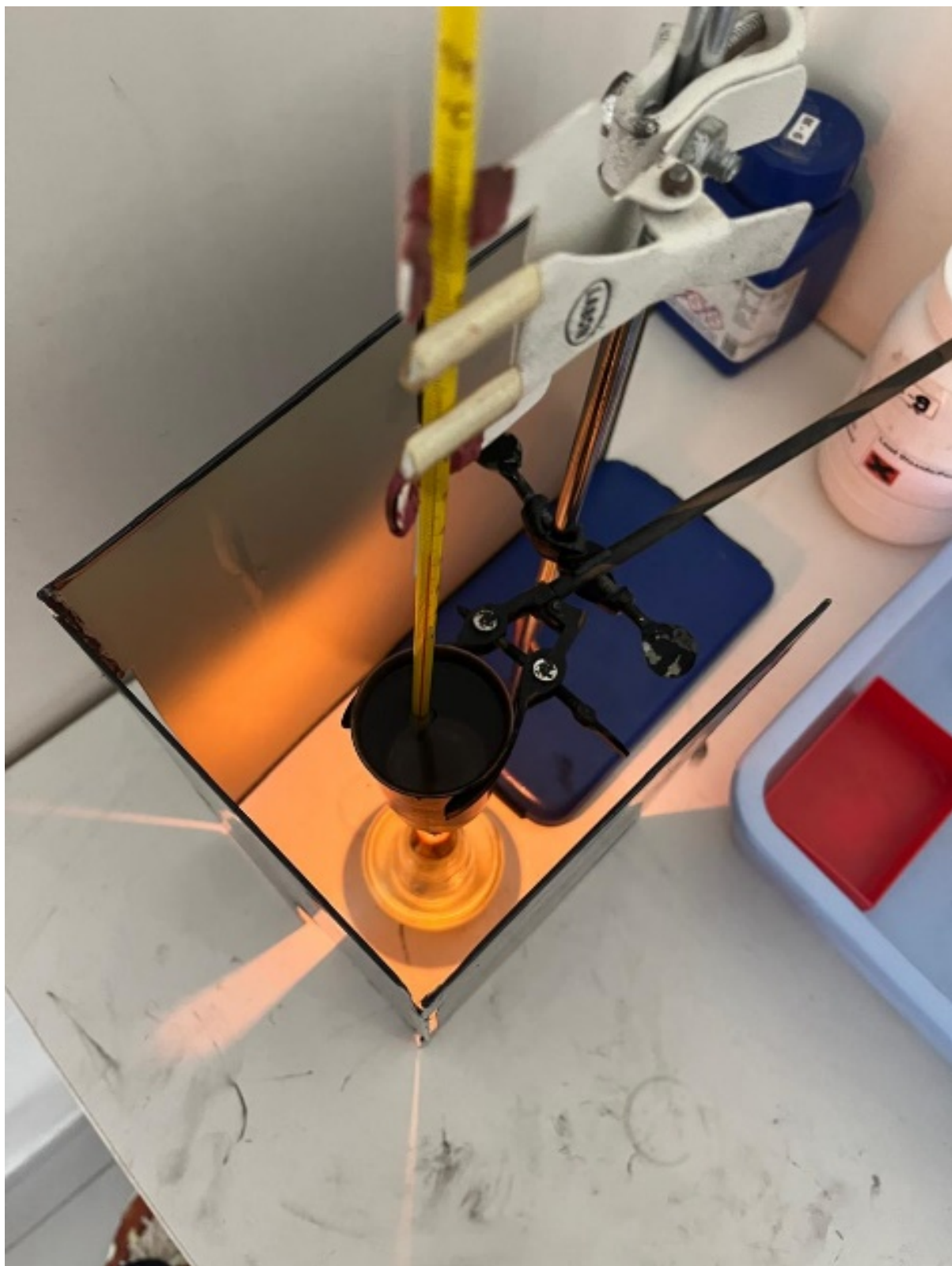


image7.jpeg

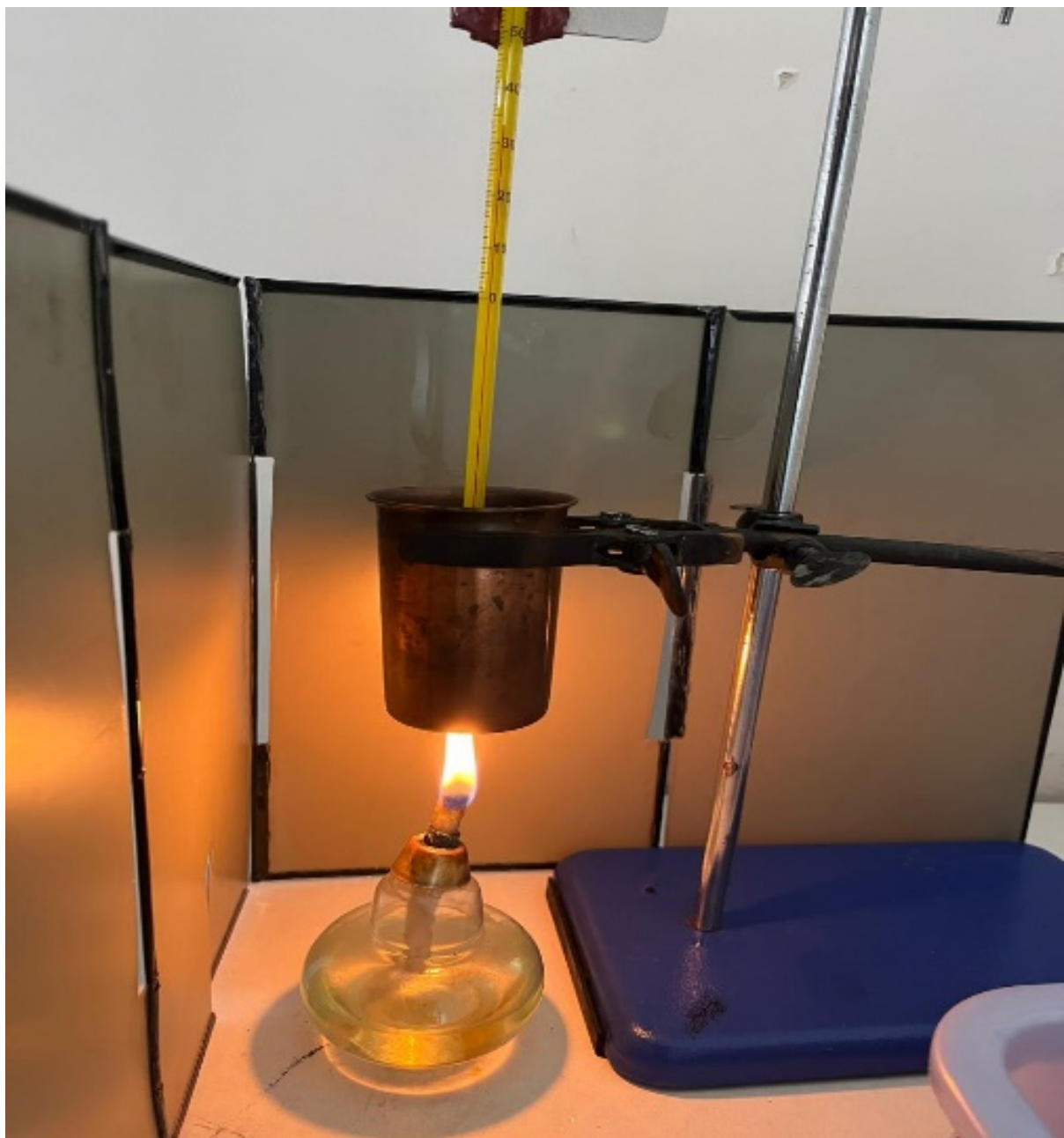


image8.png

